Product Hunt Launch Success Prediction

Final Project Report

Introduction

This project predicts Product Hunt launch success with two tasks: classification of top-20 outcomes and regression of vote counts. The approach combines structured metadata with text representations from TF-IDF and Sentence-BERT features, then evaluates linear and tree-based models.

Key challenges were API throttling and heavy-tailed vote distributions. To address these, collection used bounded windows and the regression task was trained on log-transformed targets.

Data

Source: Product Hunt GraphQL API v2. Processed samples: 1800 posts spanning years [2022, 2023, 2024, 2025]. Time range: 2022-01-29 08:00:00+00:00 to 2025-12-31 08:01:00+00:00. Unique launch days represented: 51. Top-20 class prevalence: 0.203.

Methodology

Feature sets: metadata only, metadata + TF-IDF, metadata + SBERT. Metadata columns: hour, day_of_week, month, year, maker_count, media_count, tagline_length, description_length, topic_count. TF-IDF features: 5000.

Classification uses stratified 5-fold CV with class-weighted logistic regression and regularized XGBoost variants. Regression uses 5-fold CV on log1p(votes) with LinearRegression, Ridge, XGBoost (square error), and XGBoost pseudo-Huber variants.

Implementation Details

Train/Test split sizes: 1440/360. Search spaces include model-specific depth, learning-rate, regularization, and estimator-count grids.

Best hyperparameters discovered per model:

- logreg_metadata_only: {'C': 5.0}
- logreg_metadata_tfidf: {'C': 0.1}
- logreg_metadata_sbert: {'C': 0.1}
- xgb_classifier_metadata_tfidf: {'learning_rate': 0.05, 'max_depth': 4, 'min_child_weight': 3, 'n_estimators': 150, 'reg_alpha': 0.5}
- xgb_classifier_metadata_sbert: {'learning_rate': 0.1, 'max_depth': 6, 'min_child_weight': 1, 'n_estimators': 150, 'reg_alpha': 0.0}
- ridge_metadata_tfidf: {'alpha': 10.0}
- ridge_metadata_sbert: {'alpha': 10.0}
- xgb_regressor_metadata_sbert: {'learning_rate': 0.01, 'max_depth': 7, 'n_estimators': 400, 'subsample': 0.8}
- xgb_huber_regressor_metadata_sbert: {'learning_rate': 0.01, 'max_depth': 2, 'min_child_weight': 5, 'n_estimators': 100, 'reg_lambda': 5.0, 'subsample': 0.7}

Results: Classification Metrics

model	roc_auc	f1	precision	recall

logreg_metadata_sbert	0.728	0.415	0.321	0.589
logreg_metadata_tfidf	0.726	0.392	0.314	0.521
logreg_metadata_only	0.715	0.410	0.318	0.575
xgb_classifier_metadata_sbert	0.681	0.229	0.478	0.151
xgb_classifier_metadata_tfidf	0.667	0.375	0.345	0.411

Results: Regression Metrics

model	r2	mae	r2_log
xgb_regressor_metadata_sbert	0.116	74.594	0.356
ridge_metadata_tfidf	0.020	78.649	0.280
ridge_metadata_sbert	0.004	79.366	0.272
linear_metadata_only	-0.059	80.164	0.261
daily_mean_baseline	0.054	90.357	-0.066
xgb_huber_regressor_metadata_sbert	-530.776	3279.272	-9.274

Results Summary

Best classification model: logreg_metadata_sbert (ROC-AUC=0.728). Best regression model: xgb_regressor_metadata_sbert (R2_log=0.356, R2_original=0.116, MAE=74.594).

The primary regression success criterion is log-scale R-squared because models are trained on log targets to handle skewed counts. Original-scale R-squared is reported for transparency.

Figure: Classification ROC-AUC

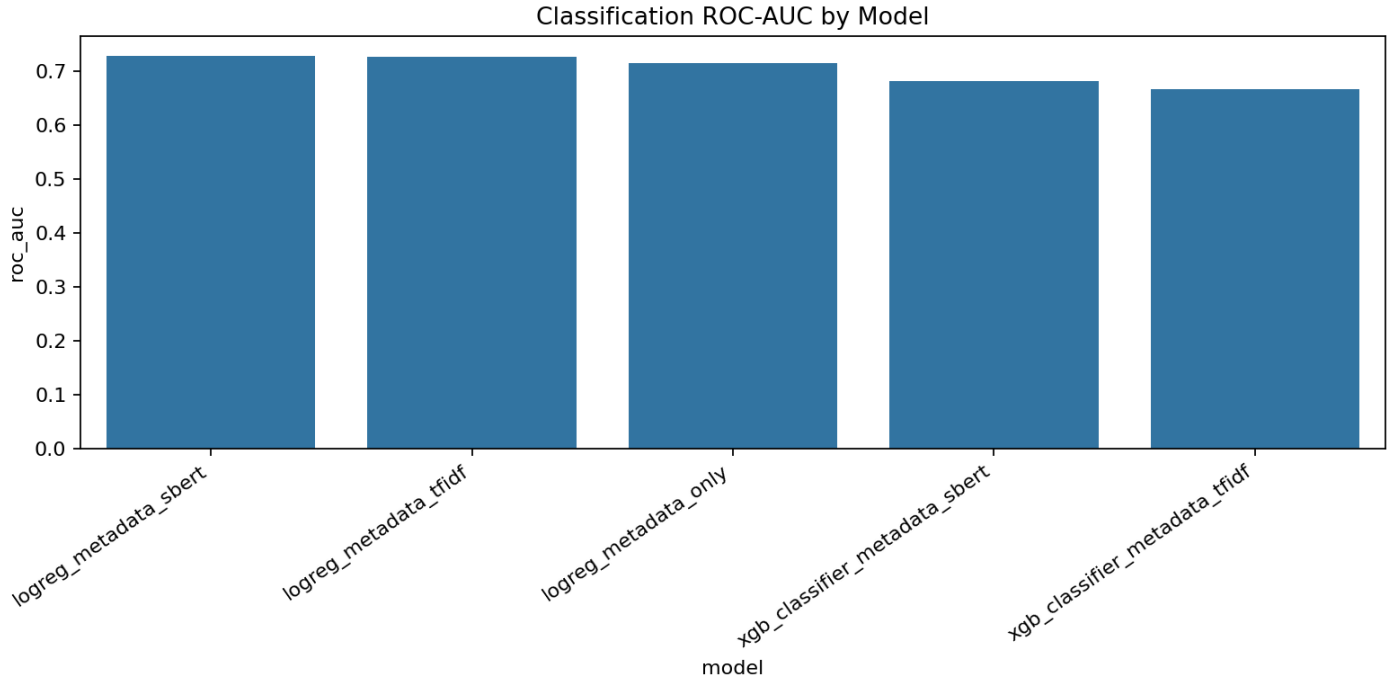

Figure: ROC Curves

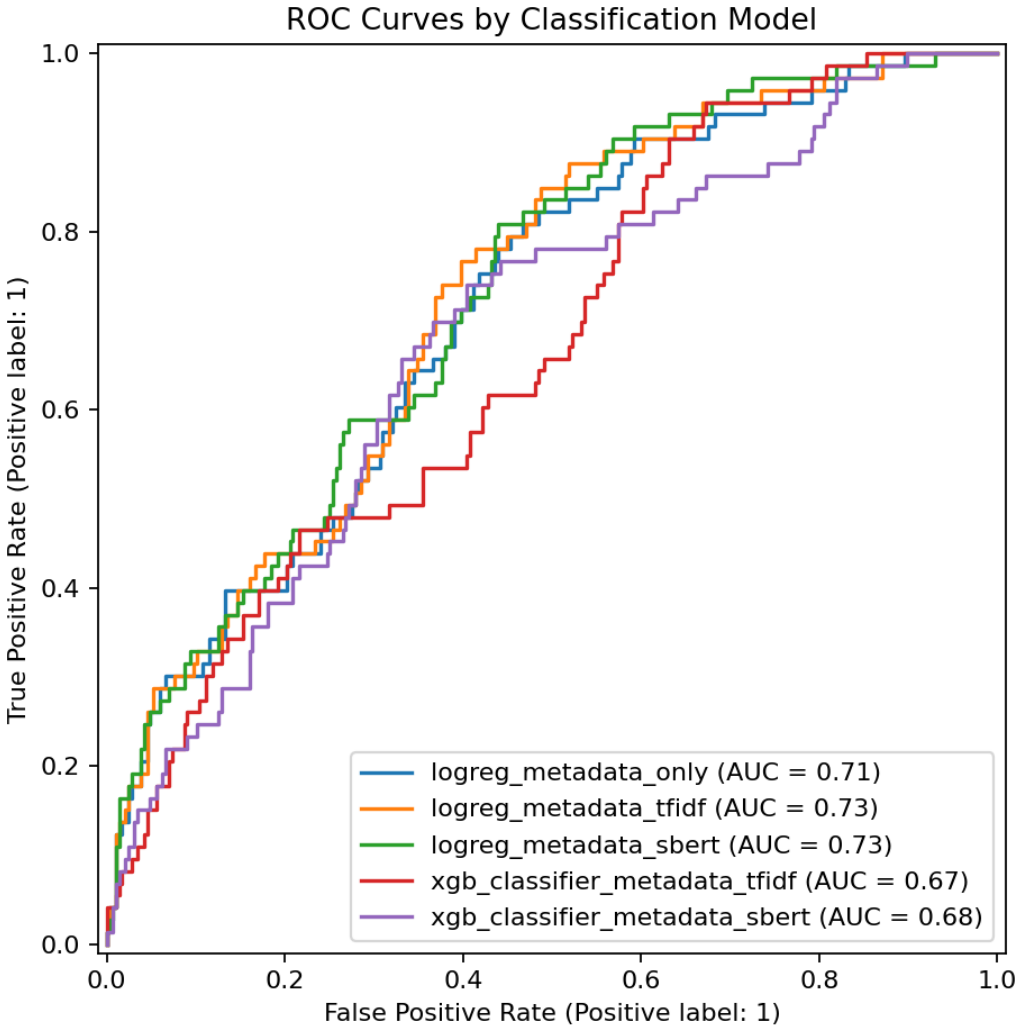

Figure: Best Confusion Matrix

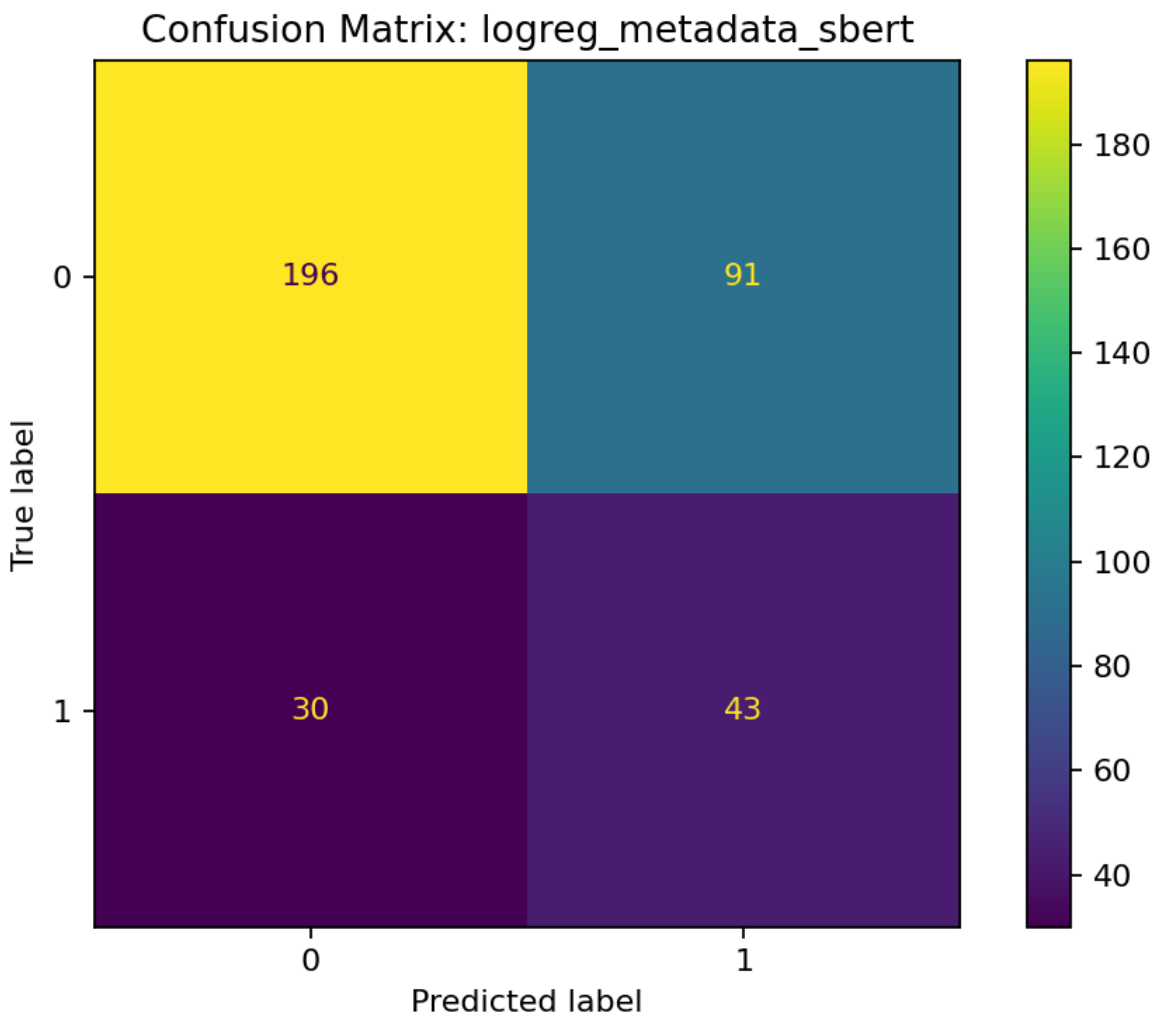

Figure: Regression Log-Scale R-squared

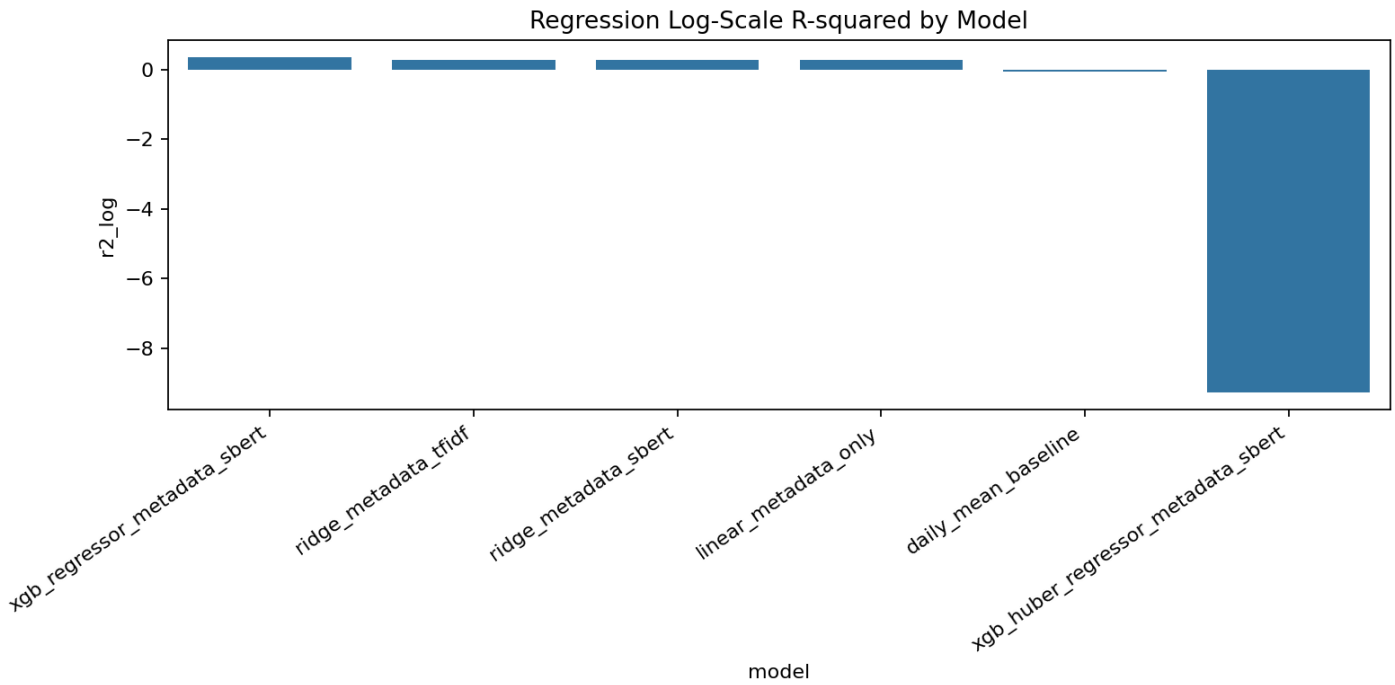

Figure: XGBoost Classifier Feature Importance

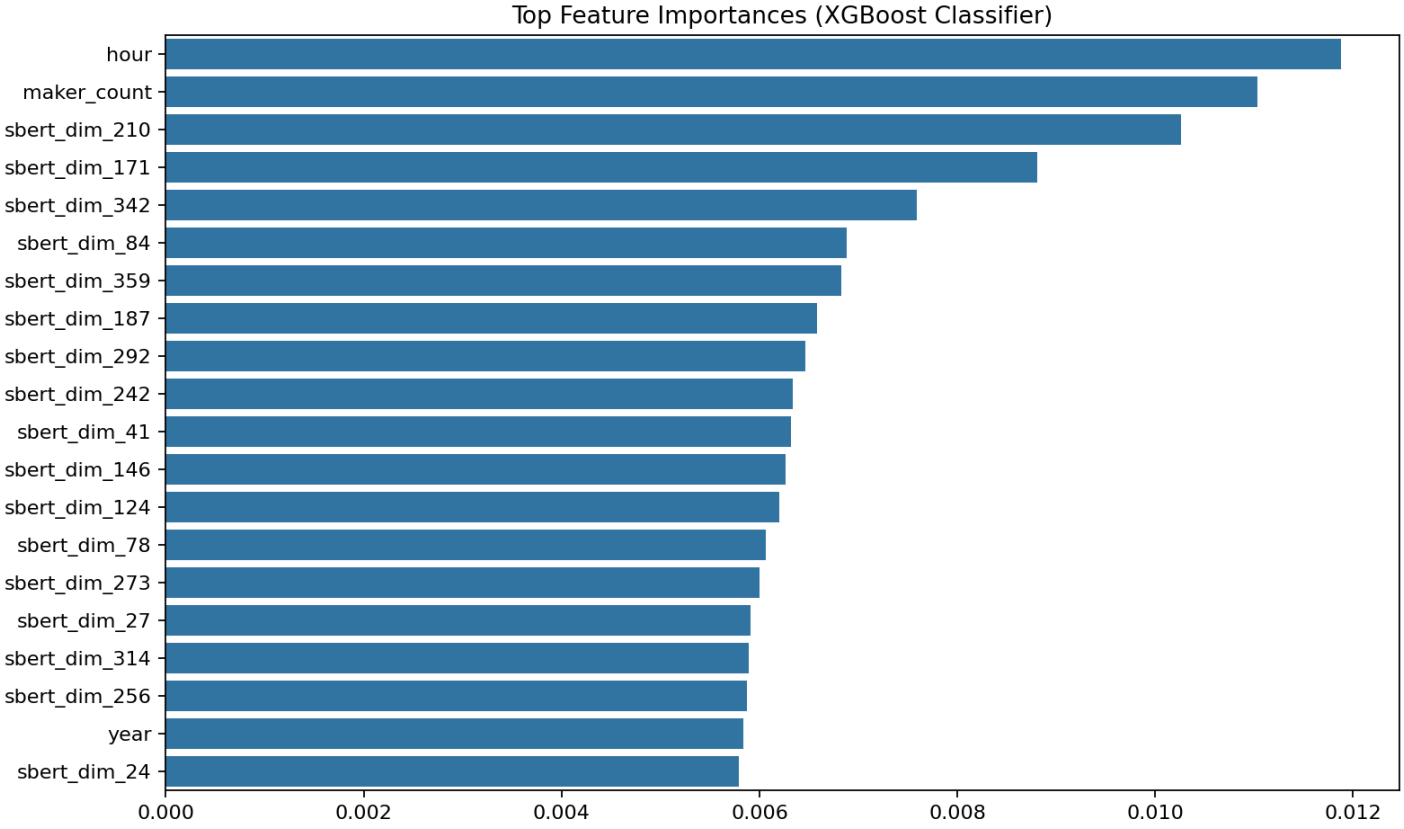
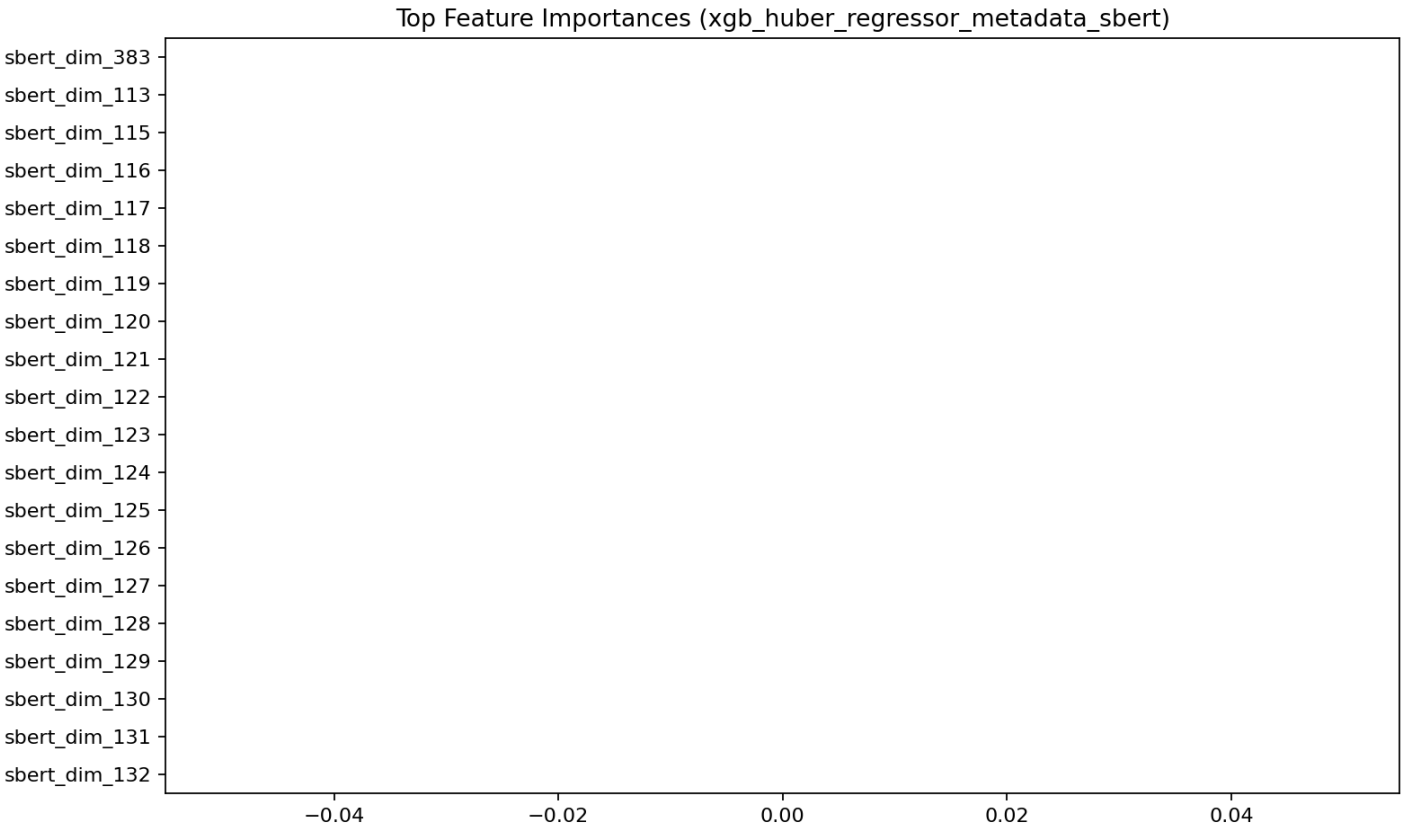
Figure: XGBoost Regressor Feature Importance